Discrete Applied Mathematics 384 (2026) 229–242

Contents lists available at ScienceDirect

Discrete Applied Mathematics

journal homepage: www.elsevier.com/locate/dam

Polarity on H-split graphs[☆]

Fernando Esteban Contreras-Mendoza, César Hernández-Cruz [*]

Facultad de Ciencias, Universidad Nacional Autónoma de Mexico, Ciudad Universitaria, Av. Universidad 3000, Ciudad de Mexico, 04510, CDMX, Mexico

ARTICLE INFO

Article history:
Received 20 June 2024
Received in revised form 23 November 2025
Accepted 30 December 2025
Available online 8 January 2026

Keywords:
Split graph
Pseudo-split graph
H-split graph
Polar graph
Degree sequence
Linear-time algorithm

ABSTRACT

A partition (A, B) of the vertex set of a graph G is called a polar partition of G when $G[A]$ and $\overline{G[B]}$ are complete multipartite graphs. If (A, B) is a polar partition of G in which $G[A]$ and $\overline{G[B]}$ have at most s and k parts, respectively, then (A, B) is an (s, k)-polar partition of G, and G is said to be an (s, k)-polar graph. A graph is said to be unipolar (monopolar) if its vertex set admits a polar partition (A, B) such that A is a clique (an independent set, respectively). A graph admitting a $(1, 1)$-polar partition is usually called a split graph.

Naturally, most problems related to polar partitions are trivial on split graphs, even when some of them are very hard in general. In this work, we present some results related to polar partitions on two graph classes generalizing split graphs. Our main results include efficient algorithms to decide whether graphs in these classes admit such partitions. We also establish upper bounds on the order of minimal (s, k)-polar obstructions for these families, for any s and k (possibly $s = \infty$ or $k = \infty$).

1. Introduction

All graphs in this work are finite and simple. In general, we follow [1], although some notation may differ slightly. As usual, we use K_n to denote the complete graph on n vertices, and $H \leq G$ to denote that the graph H is an induced subgraph of the graph G. We denote by $G+H$ and $G \oplus H$ the disjoint union and the join of the graphs G and H, respectively. Congruently, we use nG to denote the disjoint union of n copies of a graph G. Two subsets U and W of the vertex set of a graph $G = (V, E)$ are said to be *complete* (respectively, *anticomplete*) if $uw \in E$ (respectively, if $uw \notin E$) for every pair of distinct vertices $u \in U$ and $w \in W$. For a family $\mathcal{F}$ of graphs, we say that a graph G is *$\mathcal{F}$-free* if G contains no graph in $\mathcal{F}$ as an induced subgraph; if $\mathcal{F} = \{F\}$ we say that G is F-free instead of $\{F\}$-free. A property of graphs is said to be *hereditary* if it is closed under taking induced subgraphs. A *minimal $\mathcal{P}$-obstruction* for a hereditary property $\mathcal{P}$ of graphs is a graph that does not have the property $\mathcal{P}$, but such that every vertex-deleted subgraph does.

Given two nonnegative integers, s and k, a partition (A, B) of the vertex set of a graph G is called an *(s, k)-polar partition* of G if A induces a complete multipartite graph with at most s parts (i.e. a *complete s-partite graph*) and B induces the disjoint union of at most k complete graphs (which is called a *k-cluster*); an *(s, k)-polar graph* is a graph admitting an (s, k)-polar partition. If either s or k (or both) are set to ∞, it means that there is no restriction on the number of parts of $G[A]$, $\overline{G[B]}$, or both, respectively. A *polar partition* is an (∞, ∞)-polar partition. A *monopolar partition* (respectively, a *unipolar partition*) is a polar partition (A, B) such that A is an independent set (respectively, a clique); notice that a monopolar partition is exactly a $(1, \infty)$-polar partition. Naturally, graphs admitting polar, monopolar, or unipolar partitions are called

☆ The authors gratefully acknowledge support from SEP-CONACYT A1-S-8397 and UNAM-PAPIIT IN106425 grants.
* Corresponding author.
E-mail addresses: esteban.contreras@ciencias.unam.mx (F.E. Contreras-Mendoza), chc@ciencias.unam.mx (C. Hernández-Cruz).

https://doi.org/10.1016/j.dam.2025.12.065
0166-218X/

Table 1
P and NP-c indicate that the complexity status of the corresponding problem is polynomial or NP-complete, respectively.

Graph class	Monopolarity	Polarity
general graphs	NP-c [6]	NP-c [2]
chordal	$O(\|V\| + \|E\|)$ [4]	P [4]
P_5-free	$O(\|V\|^4)$ [11]	NP-c [4]
$\{2K_2, C_5\}$-free	$O(\|V\|^4)$ [4]	NP-c [4]
hole-free	P [4]	NP-c [4]
$\{C_4, C_5\}$-free planar	NP-c [4]	NP-c [4]

polar, monopolar and unipolar graphs, respectively. The $(1, 1)$-polar graphs are usually called *split graphs*. Two well-known characterizations of split graphs are given in the following proposition.

Theorem 1 ([9,10]). *Let G be a graph with vertex set $\{v_1, v_2, \dots, v_n\}$ and degree sequence $d_1 \geq d_2 \geq \cdots \geq d_n$, where d_i is the degree of vertex v_i. Set $p = \max\{i\colon\ d_i \geq i - 1\}$. The following conditions are equivalent:*

1. *G is a split graph;*
2. *G is a $\{2K_2, C_4, C_5\}$-free graph;*
3. *$\sum_{i=1}^{p} d_i = p(p-1) + \sum_{i=p+1}^{n} d_i$.*

Additionally, if G is a split graph, then $(\{v_1, \dots, v_p\}, \{v_{p+1}, \dots, v_n\})$ is a split partition of G.

Note that, once the degree sequence of a graph G is known, computing the value of p, as well as verifying the condition in item 3 of Theorem 1, can be done in $O(|V|)$ time, so split graphs are recognizable and a split partition can be found in linear time from their degree sequences.

It is known that, for any pair of nonnegative integers s and k, there are finitely many minimal (s, k)-polar obstructions [8], and (s, k)-polar graphs can be recognized in $O(|V|^{4+2\max\{s,k\}})$ time [7]. Despite this, complete lists of minimal (s, k)-polar obstructions are known only for the cases $\min\{s, k\} = 0$ and $s = k = 1$. It is well known that, for an integer $k \geq 2$, the $(0, k)$-polar graphs (k-clusters) are precisely the $\{P_3, (k+1)K_1\}$-free graphs, and the $(0, \infty)$-polar graphs, called *clusters*, are the P_3-free graphs. Analogously, if $s \geq 2$, the $(s, 0)$-polar graphs (complete s-partite graphs) coincide with the $\{\overline{P_3}, K_{s+1}\}$-free graphs, and the $(\infty, 0)$-polar graphs, better known as *complete multipartite graphs*, are exactly the $\overline{P_3}$-free graphs.

Unipolar graphs have been shown to be recognizable in $O(n^3)$ time [3,5] but, in contrast, the problems of deciding whether a graph is polar or monopolar are known to be NP-complete problems [2,6].

Clearly, split graphs are unipolar, monopolar, and (s, k)-polar graphs for any positive integers s and k, so polarity is not quite interesting when restricted to split graphs. Nevertheless, some difficult problems related to polarity have been studied in many different graph classes, including families of graphs related to split graphs for which such problems remain hard to solve. Some examples can be found in Table 1.

Considering the results mentioned above and with the purpose of finding families of graphs that contain the class of split graphs but for which polarity problems can be efficiently solved, we study two generalizations of split graphs introduced by Maffray and Preissmann [12].

Given a fixed graph H, a graph G is said to be *H-split* if V_G admits a partition (C, S, I) such that C is a clique, I is an independent set, either $S = \varnothing$ or $G[S] \cong H$, C is complete to S, and I is anticomplete to S. A partition (C, S, I) as described above is called an *H-split partition* of G. If $G = (C, S, I)$ is an H-split graph with $S \neq \varnothing$, we say that G is a *strict H-split graph*. Given a family of graphs $\mathcal{H}$, we say that G is *$\mathcal{H}$-split* if it is H-split for some $H \in \mathcal{H}$. A sequence of nonnegative integers is said to be *uniquely realizable* if, up to isomorphisms, there is only one graph having such a sequence as its degree sequence. The next theorem implies that, for a graph H whose degree sequence is uniquely realizable, the class of H-split graphs is recognizable in $O(|V|)$ time and an H-split partition also can be found in $O(|V|)$ time from their degree sequences.

Theorem 2 ([12]). *Let $d_1^* \geq \cdots \geq d_h^*$ be a uniquely realizable degree sequence and let H be its only realization. Let G be a graph with degree sequence $d_1 \geq \cdots \geq d_n$, and let q be the maximum of the set $\{i\colon\ d_i \geq i - 1 + h\}$ if it is not empty, or $q = 0$ otherwise. Then, G is an H-split graph if and only if G is a split graph or*

$$\sum_{i=1}^{q} d_i = q(q-1) + qh + \sum_{j=q+h+1}^{n} d_i$$

and $d_{q+i} = q + d_i^*$ for each $i \in \{1, \dots, h\}$. Additionally, if the condition on the degrees holds, then the sets $C = \{v_1, \dots, v_q\}$, $S = \{v_{q+1}, \dots, v_{q+h}\}$ and $I = \{v_{q+h+1}, \dots, v_n\}$ form an H-partition of G.

It is a straightforward observation that, for a fixed graph H, the class of H-split graphs is self-complementary if and only if H is, and it is hereditary if and only if either H is a split graph, in which case H-split graphs coincide with split graphs, or $H \in \{2K_2, C_4, C_5\}$. Additionally, if H is $2K_2$, C_4, or C_5, and $G = (C, S, I)$ is a strict H-split graph, then the only induced copy of H in G is $G[S]$ and the H-split partition of G is unique.

From the above observations, it is not strange that the most studied H-split graphs are the C_5-split graphs which were named *pseudo-split graphs* in [12], where it was also proved that they are precisely the $\{2K_2, C_4\}$-free graphs. Naturally, a C_5-split partition of a graph is also called a *pseudo-split partition* and, since pseudo-split graphs are strict if and only if they are not perfect, strict pseudo-split graphs are called *imperfect*.

In line with this work, we study monopolar, unipolar, and (s, k)-polar partitions on H-split graphs whenever $H \in \{C_5, 2K_2, C_4\}$. The rest of the paper is organized as follows. Section 2 is devoted to polarity on pseudo-split graphs; we provide characterizations for pseudo-split (s, k)-polar graphs, from which we derive linear-time recognition algorithms depending only on the degree sequence of the input graph. We also present complete lists of pseudo-split minimal $(1, k)$- and $(2, k)$-polar obstructions for fixed arbitrary nonnegative integer k, and we show tight upper bounds (linear in $s+k$) for the order of pseudo-split minimal (s, k)-polar obstructions. In particular, we show that there are finitely many pseudo-split minimal (s, ∞)- and (∞, k)-polar obstructions. Section 3 deals with results concerning polarity in $2K_2$-split graphs that are analogous to those given in Section 2 for pseudo-split graphs. In contrast with the results presented in Section 2, we prove that $2K_2$-split graphs admitting an (s, k)-polar partition cannot always be recognized by their degree sequence, but they still can be recognized in linear time. Also, some of the upper bounds that we present for the order of $2K_2$-split minimal (s, k)-polar obstructions are not linear in $s+k$, and we do not know whether they are tight. Since C_4-split graphs are the complements of $2K_2$-split graphs, analogous results are deduced for these graphs. Finally, in Section 4 we propose some open problems and conjectures.

2. Polarity on pseudo-split graphs

Split graphs are trivially polar, monopolar, unipolar, and (s, k)-polar graphs for any positive integers s and k. Similarly, if $G = (C, S, I)$ is an imperfect pseudo-split graph and S' is a maximum clique of $G[S]$, then a polar partition of G is given by $(C \cup (S \setminus S'), I \cup S')$; hence, pseudo-split graphs are polar graphs. In this section we study polarity properties on pseudo-split graphs. As our main results, we provide $O(|V|)$-time algorithms to decide whether a pseudo-split graph is (s, k)-polar from its degree sequence, even when one of s or k is ∞, and we prove tight upper bounds for the order of pseudo-split minimal (s, k)-polar obstructions.

It is worth noticing that C_5 admits only two essentially different polar partitions (A, B). The first is a $(2, 1)$-polar partition with A inducing P_3 and B inducing K_2; the second is a $(1, 2)$-polar partition with A inducing $\overline{K_2}$ and B inducing $\overline{P_3}$. By using the previous observation, in the next proposition we summarize some facts about the general structure of any polar partition of an arbitrary imperfect pseudo-split graph.

Lemma 3. *Let $G = (C, S, I)$ be an imperfect pseudo-split graph, and let (A, B) be a polar partition of G. Let M_C be the set of vertices in C that are anticomplete to I, and let M_I be the set of vertices in I that are complete to C.*

1. *If $G[S]$ inherits a $(2, 1)$-polar partition from (A, B), then $I \subseteq B$. Otherwise, if $G[S]$ inherits a $(1, 2)$-polar partition from (A, B), then $C \subseteq A$.*
2. *If $C \cap B \neq \varnothing$, then $G[S]$ inherits a $(2, 1)$-polar partition from (A, B). Similarly, if $I \cap A \neq \varnothing$, then $G[S]$ inherits a $(1, 2)$-polar partition from (A, B)*
3. *$C \setminus M_C \subseteq A$ and $I \setminus M_I \subseteq B$.*

Proof.

1. If $(A \cap S, B \cap S)$ is a $(2, 1)$-polar partition of $G[S]$, then, since I is anticomplete to S and A induces a $\overline{P_3}$-free graph, we have that $I \subseteq B$. The other implication can be proved similarly.
2. Let $u \in C \cap B$. Since $G[B]$ is a cluster, B induces a P_3-free graph, so $B \cap S$ is a clique and $A \cap S$ induces P_3. Therefore, $G[S]$ inherits a $(2, 1)$-polar partition from (A, B). The other implication admits an analogous proof.
3. Let $u \in C \setminus M_C$, and let $v \in I \cap N(u)$. If $u \notin A$, then $u \in B$, so $C \cap B \neq \varnothing$ and we have from items 1 and 2 that $v \in B$. But then, for any $w \in B \cap S$, $\{u, v, w\}$ induces P_3, which is impossible because B induces a cluster. Therefore, $u \in A$, so we conclude that $C \setminus M_C \subseteq A$. That $I \setminus M_I \subseteq B$ can be proved analogously. □

2.1. Recognition of pseudo-split (s, k)-polar graphs

Lemma 3 reveals that the sets M_C and M_I are the key to determine polar partitions of an imperfect pseudo-split graph. In this section, we use such a property to completely characterize pseudo-split graphs admitting unipolar, monopolar, and (s, k)-polar partitions. From these characterizations we derive efficient algorithms to recognize pseudo-split graphs having the mentioned partitions.

We begin by pointing out a direct consequence of Theorem 2. Notice that, in the notation of Lemma 3, this observation implies that, for any pseudo-split graph $G = (C, S, I)$, the cardinalities of C, I, M_C, and M_I, can be computed in $O(|V|)$ time from the degree sequence of G.

Remark 4. Let $G = (C, S, I)$ be an imperfect pseudo-split graph. For each $u \in V_G$, $\mathrm{d}(u) = |C| + 4$ if and only if u is a vertex in C that is anticomplete to I, and $\mathrm{d}(u) = |C|$ if and only if u is a vertex in I that is complete to C.

It can be easily verified that C_5 is a minimal unipolar obstruction and, since split graphs are unipolar, it follows that a pseudo-split graph is unipolar if and only if it is a split graph, which can be checked in $O(|V|)$ time from its degree sequence. Note that a perfect pseudo-split graph is a split graph and hence an (s, k)-polar graph for any fixed positive integers s and k. Now, we present a necessary and sufficient condition for a pseudo-split graph to be (s, k)-polar for any fixed nonnegative integers s and k with $s + k \geq 1$.

Theorem 5. *Let $G = (C, S, I)$ be an imperfect pseudo-split graph, and let s and k be nonnegative integers. Let M_C be the set of vertices in C that are anticomplete to I, and M_I be the set of vertices in I that are complete to C. Then, G is an (s, k)-polar graph if and only if either*

1. *$k \geq |I| + 1$ and $s \geq |C| - |M_C| + 2$, or*
2. *$s \geq |C| + 1$ and $k \geq |I| - |M_I| + 2$.*

Proof. Let $c = |C|$, $i = |I|$, $m_c = |M_C|$, and $m_i = |M_I|$. First, let us suppose that G admits an (s, k)-polar partition (A, B). If $G[S]$ inherits a $(2, 1)$-polar partition from (A, B), we have from Lemma 3 that $I \subseteq B$ and $C \cap B \subseteq M_C$. Thus,

$$|C \cap A| = |C| - |C \cap B| \geq c - m_C,$$

where we conclude that $s \geq c - m_C + 2$. Furthermore, in this case $G[B]$ is a cluster with exactly $i + 1$ components, so $k \geq i + 1$. Hence, we have proved that, if $G[S]$ inherits a $(2, 1)$-polar partition from (A, B), then $k \geq i + 1$ and $s \geq c - m_C + 2$. If $G[S]$ does not inherit a $(2, 1)$-polar partition form (A, B), then it inherits a $(1, 2)$-polar partition, and it can be proved using analogous arguments that in such a case $s \geq c + 1$ and $k \geq i - m_I + 2$.

Conversely, let us assume that $k \geq i + 1$ and $s \geq c - m_C + 2$. Let B_1 be a maximum clique of $G[S]$, and let $A_1 = S \setminus B_1$. Then, we have that $(A_1 \cup (C \setminus M_C), B_1 \cup M_C \cup I)$ is a $(c - m_C + 2, i + 1)$-polar partition of G, so G is an (s, k)-polar graph, as we had to prove. The result follows analogously if we assume that $s \geq c + 1$ and $k \geq i - m_I + 2$, only taking a $(1, 2)$-polar partition (A_1, B_1) of $G[S]$ instead of a $(2, 1)$-polar partition. □

Notice that, by Remark 4, the conditions in Theorem 5 can be verified in $O(|V|)$ time from the degree sequence of the input graph, so (s, k)-polarity can be decided in linear time on this family of graphs, and an (s, k)-polar partition can be found with the same time complexity.

Next, we characterize pseudo-split graphs admitting an (s, ∞)-polar partition, where s is any fixed nonnegative integer. Once again, it follows from Remark 4 that the conditions in Theorem 6 can be verified in $O(|V|)$ time from the degree sequence of any pseudo-split graph, so (s, ∞)-polarity can be decided in linear time on pseudo-split graphs, and an (s, ∞)-polar partition can be found with the same complexity. Since monopolar graphs are precisely the $(1, \infty)$-polar graphs, the same is true for monopolarity on pseudo-split graphs.

Theorem 6. *Let s be a nonnegative integer. If $G = (C, S, I)$ is an imperfect pseudo-split graph, then G is an (s, ∞)-polar graph if and only if either $s \geq |C| + 1$, or $|C| \geq s \geq 2$ and there are at least $|C| - s + 2$ vertices in C that are anticomplete to I.*

Proof. Let us denote $|C|$ by c. Suppose that (A, B) is an (s, ∞)-polar partition of G such that $s \leq c$. Observe that, by Lemma 3, if the restriction of (A, B) to S is a $(1, 2)$-polar partition, then $C \subseteq A$, but that is impossible since $G[A]$ would have K_{c+1} as an induced subgraph and $s \leq c$. Thus, $G[S]$ is covered by a $(2, 1)$-polar partition, and we have from Lemma 3 that $I \subseteq B$.

Since A induces a complete s-partite graph and $G[A \cap S]$ is a complete bipartite graph, at most $s - 2$ vertices of C belong to A. Then, at least $c - s + 2$ vertices of C belong to B, and we have from Lemma 3 that any such vertex is anticomplete to I. This proves that, if G is (s, ∞)-polar, then it satisfies the stated conditions on c and s.

For the converse implication, let A and B be a maximum independent set and a maximum clique of $G[S]$, respectively. If $c < s$, then $(C \cup A, I \cup (S \setminus A))$ is an (s, ∞)-polar partition of G. Otherwise, we have that $c \geq s$ and the subset C' of C of all vertices that are anticomplete to I has at least $c - s + 2$ elements. Then, $((S \setminus B) \cup (C \setminus C'), I \cup C' \cup B)$ is an (s, ∞)-polar partition of G. □

Since pseudo-split graphs form a self-complementary class of graphs, and a graph G is (s, ∞)-polar if and only if $\overline{G}$ is (∞, s)-polar, we have, by a simple argument of complements, that a characterization analogous to Theorem 6 can be easily stated for (∞, k)-polarity on pseudo-split graphs.

Now, we present some results about pseudo-split minimal (s, k)-polar obstructions, which include tight upper bounds for the order of such graphs, as well as complete lists of minimal forbidden induced subgraphs for some small values of s and k.

2.2. Pseudo-split minimal (s, k)-polar obstructions

As we mentioned before, minimal (s, k)-polar obstructions on general graphs are known only for the cases $\min\{s, k\} = 0$, and $s = k = 1$, which correspond to clusters, complete multipartite graphs, and split graphs. In the following proposition we give complete lists of pseudo-split minimal (s, k)-polar obstructions for the case $s \in \{1, 2\}$, which can be easily extrapolated to the case $k \in \{1, 2\}$ by a simple complementation argument. Before presenting such results, we introduce some particular graphs.

Let $G_s^0 = (C, S, I)$ be the imperfect pseudo-split graph such that $|C| = s$, $|I| = 1$, and C is complete to I. Let G_s^1 be the graph obtained from G_s^0 by deleting one edge incident with the only vertex of I. By using Theorem 5, it is easy to verify that G_s^0 and G_s^1 are minimal (s, k)-polar obstructions whenever s and k are integers, $s, k \geq 2$.

Proposition 7. *Let k be an integer, $k \geq 2$. If $G = (C, S, I)$ is a pseudo-split graph, then*

1. *G is a $(1, k)$-polar graph if and only if G is a $\overline{K_1 \oplus C_5}$-free graph.*
2. *G is a $(2, k)$-polar graph if and only if G is a $\{G_2^0, G_2^1, \overline{G_k^0}, \overline{G_k^1}\}$-free graph.*

Proof. It follows from Theorem 5 that $K_1 \oplus C_5$ is a minimal $(1, k)$-polar obstruction. Conversely, G is $K_1 \oplus C_5$-free if and only if $S = \varnothing$ or $C = \varnothing$, but in both cases G is a $(1, 2)$-polar graph, hence a $(1, k)$-polar graph. This proves the first item.

Previously, we observed that, if s and k are integers, $s, k \geq 2$, then the graphs G_s^0 and G_s^1 are minimal (s, k)-polar obstructions. Thus, in particular, $G_2^0, G_2^1, \overline{G_k^0}$, and $\overline{G_k^1}$ are minimal $(2, k)$-polar obstructions for any $k \geq 2$, so no $(2, k)$-polar graph has any of these graphs as an induced subgraph.

Now, assume that G is a $\{G_2^0, G_2^1, \overline{G_k^0}, \overline{G_k^1}\}$-free graph. If $S = \varnothing$, then G is a split graph, hence a $(2, k)$-polar graph. Additionally, if either $C = \varnothing$ or $I = \varnothing$, then G trivially is $(1, 2)$- or $(2, 1)$-polar, respectively, so, in any of these cases, G is a $(2, k)$-polar graph. Therefore, we can assume that G is an imperfect pseudo-split graph such that $|C|, |I| \geq 1$.

Notice that, if C is anticomplete to I, then $(S \setminus S', C \cup S' \cup I)$ is a $(2, |I| + 1)$-polar partition of G for any maximum clique S' of $G[S]$. In particular, if $|C| \geq 2$, since G is a $\{G_2^0, G_2^1, \overline{G_k^0}\}$-free graph, we have that C is completely nonadjacent to I, and that $|I| \leq k - 1$. Thus, by the initial observation in this paragraph, G is a $(2, k)$-polar graph.

Finally, we can assume now that C has only one vertex, say v. Since G is $\{\overline{G_k^0}, \overline{G_k^1}\}$-free, v is adjacent to at least $|I| - k + 2$ vertices in I, and then a $(2, k)$-polar partition of G is given by $(C \cup S' \cup (N(v) \cap I), (S \setminus S') \cup (I \setminus N(v)))$, where S' is a maximum independent set in $G[S]$. $\square$

It is a simple observation that, for any nonnegative integer s, a graph G is a minimal (s, ∞)-polar obstruction if and only if there is a nonnegative integer k_0 such that, for any integer $k \geq k_0$, G is a minimal (s, k)-polar obstruction. Then, we have the following immediate consequence of Proposition 7.

Proposition 8. *A pseudo-split graph G is $(1, \infty)$-polar (i.e. monopolar) if and only if it is $K_1 \oplus C_5$-free, and it is $(2, \infty)$-polar if and only if it is a $\{G_2^0, G_2^1\}$-free graph.*

It seems that there is not an easy way to describe the complete lists of pseudo-split minimal (s, k)-polar obstructions when s and k are any fixed nonnegative integers, but, as we mentioned before, we know that there is just a finite number of them, so it becomes natural to ask about the best possible upper bounds for their order. In the following propositions, we use Theorem 5 to give tight upper bounds for the order of pseudo-split minimal (s, k)-polar obstructions. We start with the following technical observation.

Lemma 9. *Let s and k be nonnegative integers. If $G = (C, S, I)$ is an imperfect pseudo-split graph, then*

1. *If $|C| > s$, then G is not a minimal (s, k)-polar obstruction.*
2. *If $|I| > k$, then G is not a minimal (s, k)-polar obstruction.*

Proof. We only prove item 1 because the proof of item 2 is analogous. Let $c = |C|$ and $i = |I|$, and assume that $c > s$. If $i \geq k$, it follows from Theorem 5 that, for every vertex $v \in C$, $G - v$ is not an (s, k)-polar graph, so G is not a minimal (s, k)-polar obstruction. Thus, we can assume that $i < k$.

Now, assume that G is not an (s, k)-polar graph. We have from Theorem 5 that at most $c - s + 1$ vertices in C are anticomplete to I, and then, at least $s - 1$ vertices in C have some neighbor in I. Let C' any s-subset of C containing at least $s - 1$ vertices having at least one neighbor in I, and let $H = G[C' \cup S \cup I]$. Clearly, H is a proper induced subgraph of G, and we have from Theorem 5 that H does not admit an (s, k)-polar partition. Thus, G is not a minimal (s, k)-polar obstruction. $\square$

By itself, Lemma 9 implies that, for any nonnegative integers s and k, each pseudo-split minimal (s, k)-polar obstruction has order at most $s + k + 5$. Nevertheless, as we can conclude from Proposition 7 and the observations that precede it, if $\min\{s, k\} \leq 2$, each minimal (s, k)-polar obstruction has order strictly smaller than $s + k + 5$. In Lemma 10 and Theorem 11 we will prove that the same is true for general values of s and k.

Lemma 10. *If s and k are integers, $s, k \geq 2$, then any pseudo-split minimal (s, k)-polar obstruction $G = (C, S, I)$ is imperfect, and satisfies that $|C| \leq s$, $|I| \leq k$, and $|C \cup I| \leq s + k - 1$. Consequently, any pseudo-split minimal (s, k)-polar obstruction has order at most $s + k + 4$, and this bound is tight when $\min\{s, k\} = 2$.*

Proof. Let $G = (C, S, I)$ be a pseudo-split minimal (s, k)-polar obstruction. Split graphs are $(1, 1)$-polar, hence (s, k)-polar, so G is an imperfect pseudo-split graph. Let $c = |C|$ and $i = |I|$. Observe that Lemma 9 implies that $s \geq c$, $k \geq i$, and that $|V_G| = s + k + 5$ if and only if $c = s$ and $i = k$.

By way of contradiction, assume that $c = s$ and $i = k$. Let $v \in C$, and let $C' = C \setminus \{v\}$. Let (A, B) be an (s, k)-polar partition of $G - v$. It is impossible that $I \subseteq B$, because $S \cap B \neq \varnothing$ and B induces a $(k + 1)K_1$-free graph. Thus, there is a vertex $u \in I \cap A$, and we have from Lemma 3 that u is complete to C'. Here we have the desired contradiction, because $G[C \cup S \cup \{u\}]$ is isomorphic to either G^0_s or G^1_s, depending on whether u is adjacent to v or not, but then G has an (s, k)-polar obstruction as a proper induced subgraph, contradicting the minimality of G. The contradiction arose from assuming that $|V_G| > s + k + 4$, so this is not the case. Notice that G^0_s and G^1_s attain the bound whenever $k = 2$, so the bound is tight whenever $\min\{s, k\} = 2$. □

As we show in the following theorem, the upper bound given in Lemma 10 is not the best possible whenever $s, k > 2$, but by making a small adjustment, we can make it tight.

Theorem 11. *If s and k are integers, $s, k \geq 3$, then any pseudo-split minimal (s, k)-polar obstruction has order at most $s+k+3$, and the bound is tight.*

Proof. Let $G = (C, S, I)$ be a pseudo-split minimal (s, k)-polar obstruction, and let c and i be the cardinalities of C and I, respectively. By Lemma 10, G is an imperfect pseudo-split graph with $c \leq s$, $i \leq k$ and, either $c \leq s - 1$ or $i \leq k - 1$. Notice that, if $c < s - 1$ or $i < k - 1$, then $|V_G| \leq s + k + 3$, so we are done. Thus, we can assume that, either $c = s - 1$ or $i = k - 1$. We focus on the case $i = k - 1$, the case $c = s - 1$ is analogous.

By way of contradiction, let us suppose that G has at least $s+k+4$ vertices, which implies by the previous observations that $c = s$. Let v be a vertex in C, and let (A, B) be an (s, k)-polar partition of $G - v$.

Notice that, if there is a vertex $u \in A \cap I$, it follows from Lemma 3 that u is complete to $C \setminus \{v\}$, but then $G[C \cup S \cup \{u\}]$ is isomorphic to either G^0_s or G^1_s, so G properly contains an (s, k)-polar obstruction, which is impossible. Therefore, for any $v \in C$ and any (s, k)-polar partition (A, B) of $G - v$, we have $I \cap A = \varnothing$.

In particular, it must be the case that $I \subseteq B$, which implies that $G[S]$ inherits a $(2, 1)$-polar partition (A', B') from (A, B), so there is a vertex $w \in B \cap (C \setminus \{v\})$ which, by Lemma 3, is anticomplete to I. Similarly, there is a vertex $w' \in C \setminus w$ that is anticomplete to I. But then,

$$(A' \cup C \setminus \{w, w'\}, B' \cup I \cup \{w, w'\})$$

is an (s, k)-polar partition of G, which is absurd. The contradiction arose from supposing that $|V_G| \geq s + k + 4$, so it must be the case that G has at most $s + k + 3$ vertices.

Now, for any integers s and k such that $k \geq s \geq 3$, let $H^k_s = (C, S, I)$ be the imperfect pseudo-split graph such that $|C| = s - 1$, $|I| = k - 1$ and, for an injection $f : C \to I$, a vertex $v \in C$ is adjacent to a vertex $u \in I$ if and only if $u = f(v)$. Notice that, as a consequence of Theorem 5, H^k_s is a minimal (s, k)-polar obstruction of order $s + k + 3$, so the bound is tight. □

In contrast with minimal (s, k)-polar obstructions when s and k are integers, it is unknown whether the number of minimal (s, ∞)-polar obstructions for general graphs is finite. In the following propositions we prove that, restricted to the class of pseudo-split graphs, all minimal (s, ∞)-polar obstructions are minimal $(s, s + 1)$-polar obstructions, implying that there is only a finite number of them. We start with some technical propositions.

For each integer $s \geq 2$, let $F_s = (C, S, I)$ be the imperfect pseudo-split graph such that $|C| = s$, $|I| = s - 1$ and, for an injection $f : I \to C$, a vertex $v \in I$ is adjacent to a vertex $u \in C$ if and only if $u = f(v)$. Notice that, it follows from Theorem 5 that F_s is not an (s, ∞)-polar graph, and it is a minimal (s, k)-polar obstruction if and only if $k \geq s + 1$.

Lemma 12. *Let s and k be integers, $s, k \geq 3$. If $G = (C, S, I)$ is a pseudo-split minimal (s, k)-polar obstruction with $|C| = s$, then $|I| \leq s - 1$. In addition, if $|I| = s - 1$, then $G \cong F_s$.*

Proof. Let $c = |C|$ and $i = |I|$. For each $v \in I$, $G - v$ has an (s, k)-polar partition (A, B), and we have from Lemma 3 that, since $c = s$, then $I \setminus \{v\} \subseteq B$ and at least two vertices in C are anticomplete to $I \setminus \{v\}$.

Now, it follows from Theorem 5 that there is at most one vertex of C that is anticomplete to I. Hence, for each vertex $v \in I$, there is a vertex $u \in C$ whose only neighbor in I is v. Thus, if $i \geq s - 1$, then F_s is an induced subgraph of G. From here, it easily follows that $i \leq s - 1$ and $G \cong F_s$ whenever $i = s - 1$. □

Lemma 13. *Let s and k be integers, $k > s \geq 3$. If $G = (C, S, I)$ is a pseudo-split minimal (s, k)-polar obstruction such that $|C| = s$, then G is a minimal (s, k^*)-polar obstruction for every integer $k^* \geq s + 1$.*

Proof. From Lemma 12 we have that either $G \cong F_s$ or $|I| < s - 1$. Moreover, as we previously observed, F_s is a minimal (s, k^*)-polar obstruction whenever $k^* \geq s + 1$, so in that case the result follows. Thus, for the rest of this proof we assume that $|I| \leq s - 2$. We will prove that G is not an (s, ∞)-polar graph and that every vertex-deleted subgraph of G is $(s, s + 1)$-polar, which implies the desired result.

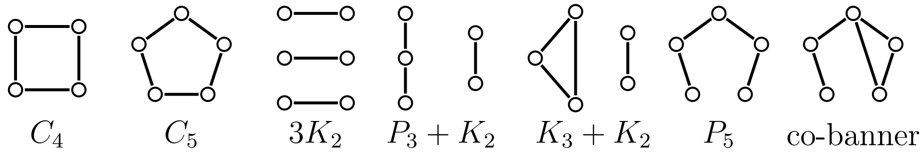

Fig. 1. Minimal $2K_2$-split obstructions.

Since G is not an (s,k)-polar graph and $|I| \leq s-2 < k-2$, it follows from item 1 of Theorem 5 that at most one vertex in C is anticomplete to I. Because of this and since $|C| = s$, Theorem 5 implies that G is not an (s,∞)-polar graph.

Let v be a vertex of G. If $v \in S$, then $G-v$ is a split graph, hence an $(s,s+1)$-polar graph. If $v \in C$, then it follows from item 2 of Theorem 5 that $G-v$ is $(s,s+1)$-polar. Now, assume that $v \in I$. Since G is a minimal (s,k)-polar obstruction, $G-v$ is an (s,k)-polar graph, and Theorem 5 implies that at least two vertices of C are adjacent to $I \setminus \{v\}$. Thus, we have from item 1 of Theorem 5 that $G-v$ is an $(s,s+1)$-polar graph. $\square$

Theorem 14. *Let s be an integer, $s \geq 3$. Any pseudo-split minimal (s,∞)-polar obstruction has order at most $2s+4$, and the bound is tight. In consequence, there are finitely many minimal (s,∞)-polar obstructions.*

Proof. It is clear from Lemma 13 that, if $G = (C,S,I)$ is a minimal $(s,s+1)$-polar obstruction with $|C| = s$, then G is a minimal (s,∞)-polar obstruction. We start by proving that the converse implication is also true.

Let $G = (C,S,I)$ be a pseudo-split minimal (s,∞)-polar obstruction, and let $c = |C|$ and $i = |I|$. Notice that, by Lemma 13, G is a minimal $(s,s+1)$-polar obstruction, so it only remains to prove that $c = s$.

By hypothesis, G is not an (s,∞)-polar graph, so G is not $(s,i+2)$-polar. Thus, we have from Theorem 5 that $s \leq c$ and there is at most one vertex in C that is anticomplete to I.

By way of contradiction, assume that $c > s$, and let $v \in C$. By the minimality of G, the graph $G-v$ is (s,k)-polar for some nonnegative integer k. Then, since $c > s$, Theorem 5 implies that at least two vertices of $C \setminus \{v\}$ are anticomplete to I, contrary to our previous observation. Since it is impossible that $c > s$, we conclude that $c = s$.

By the previous arguments we have that, if $G = (C,S,I)$ is a pseudo-split minimal (s,∞)-polar obstruction, then G is a minimal $(s,s+1)$-polar obstruction with $|C| = s$. Then, we have from Theorem 11 that the order of G is at most $2s+4$. This bound is attained by F_s. $\square$

Throughout this section, we gave linear-time algorithms for deciding whether a pseudo-split graph admits a monopolar, unipolar, or (s,k)-polar partition, and we provided tight upper bounds for the order of the minimal obstructions associated with such partitions. In the next section, we develop analogous results for $2K_2$- and C_4-split graphs.

3. $2K_2$- and C_4-split graphs

Theorem 2 provide us with a characterization of $2K_2$-split graphs that generalizes item 3 of Theorem 1. Next, we give a characterization of $2K_2$-split graphs that is analogous to Item 2 of Theorem 1 for split graphs. Notice that the minimal obstructions are C_4, C_5, every possible way of adding one new vertex to $2K_2$ such that it is adjacent to at least one (but not all) of its vertices, and $3K_2$, which is obtained from $2K_2$ by adding two new vertices which are adjacent only to each other.

Theorem 15. *A graph G is a $2K_2$-split graph if and only if G has no induced subgraphs isomorphic to the graphs depicted in Fig. 1.*

Proof. It is a routine job to check that any graph in Fig. 1 is a minimal $2K_2$-split obstruction. Conversely, let G be a graph without induced subgraphs isomorphic to the graphs in Fig. 1. If G is $2K_2$-free, then G is a split graph, and hence a $2K_2$-split graph, so let us assume that G has an induced copy of $2K_2$ with vertex set S. Notice that any vertex in $V_G \setminus S$ is either completely adjacent to S or anticomplete to it, or G would have some of the forbidden induced subgraphs.

Since G is a $3K_2$-free graph, any two vertices $u, v \in V_G \setminus S$ that are anticomplete to S must be nonadjacent to each other. In addition, if there exist two nonadjacent vertices $u, v \in V_G \setminus S$ that are complete to S, then G has an induced C_4, which is impossible.

Hence, if we set C to be the set of all vertices in $V_G \setminus S$ that are complete to S, and I the set of vertices in $V_G \setminus S$ that are anticomplete to S, then (C,S,I) is a $2K_2$-split partition of G. $\square$

In this section we study polarity on $2K_2$-split graphs. Observe that a graph is $2K_2$-split if and only if its complement is C_4-split. Thus, since the complement of an (s,k)-polar graph is a (k,s)-polar graph, by simple complementation arguments, any result about (s,k)-polarity on $2K_2$-split graphs is equivalent to a dual (k,s)-polarity result on C_4-split graphs. As the reader will notice, although some results are similar to those proved in Section 2 for pseudo-split graphs, there are also remarkable differences.

3.1. $2K_2$-split minimal (s, k)-polar obstructions

As in the case of pseudo-split graphs, any $2K_2$-split graph is polar. Moreover, if $G = (C, S, I)$ is a $2K_2$-split graph, then $(C, S \cup I)$ is a unipolar partition of G, so G is unipolar, and hence polar. Similarly, any C_4-split graph $G = (C, S, I)$ has a unipolar partition, namely $(C \cup S', I \cup (S \setminus S'))$, where S' is any maximum clique of $G[S]$. Thus, any C_4-split graph is unipolar.

A relevant difference between pseudo-split and $2K_2$-split graphs is that C_5 admits only two essentially distinct polar partitions, while $2K_2$ admits four such partitions. By using that complete multipartite graphs are $\overline{P_3}$-free, it can be easily verified that one of the following conditions must occur for each polar partition (A, B) of $2K_2$:

- T_1 partition: A induces K_2, so (A, B) is a $(2, 1)$-polar partition.
- T_2 partition: A induces $2K_1$, so (A, B) is a $(1, 2)$-polar partition.
- T_3 partition: A induces K_1, so (A, B) is a $(1, 2)$-polar partition.
- T_4 partition: $A = \varnothing$, so (A, B) is a $(0, 2)$-polar partition.

The next lemma relates to $2K_2$-split graphs the same as Lemma 3 relates to pseudo-split graphs. In it, we state some properties of polar partitions of strict $2K_2$-split graphs depending on the partition that its only copy of $2K_2$ inherits. We omit the proof of this lemma because it is quite similar to that of Lemma 3.

Lemma 16. *Let $G = (C, S, I)$ be a strict $2K_2$-split graph, and let (A, B) be a polar partition of G. Let M_C be the set of vertices in C that are anticomplete to I, and let M_I be the set of vertices in I that are complete to C.*

1. *If $G[S]$ inherits a T_1 partition from (A, B), then $I \subseteq B$, and $C \setminus M_C \subseteq A$.*
2. *If $G[S]$ inherits either a T_2 or a T_3 partition from (A, B), then $C \subseteq A$, and $I \setminus M_I \subseteq B$.*
3. *If $G[S]$ inherits a T_4 partition from (A, B), then $C \subseteq A$.*

In consequence,

1. *If $C \cap B \neq \varnothing$, then $G[S]$ inherits a T_1 partition from (A, B), and $M_C \neq \varnothing$.*
2. *If $I \cap A \neq \varnothing$, then $G[S]$ inherits a T_2, T_3, or T_4 partition from (A, B).*
3. *If $(I \setminus M_I) \cap A \neq \varnothing$, then $G[S]$ inherits a partition of type T_4 from (A, B).*

The following proposition, completely characterizes the $2K_2$-split and C_4-split graphs that admit a $(1, k)$-polar partition for any integer $k \geq 2$.

Proposition 17. *Let k be an integer, $k \geq 2$.*

1. *A $2K_2$-split graph $G = (C, S, I)$ is a $(1, k)$-polar graph if and only if G is a $\{K_2 \oplus 2K_2, K_1 \oplus (2K_2 + \overline{K_{k-1}})\}$-free graph. In consequence, the only $2K_2$-split minimal monopolar obstruction is $K_2 \oplus 2K_2$.*
2. *A C_4-split graph $G = (C, S, I)$ is a $(1, k)$-polar graph if and only if G is a $K_1 \oplus C_4$-free graph. Consequently, the only C_4-split minimal monopolar obstruction is $K_1 \oplus C_4$.*

Proof. We only prove the first statement, because the second one admits an analogous proof. By using Lemma 16, it is easy to verify that both, $K_2 \oplus 2K_2$ and $K_1 \oplus (2K_2 + \overline{K_{k-1}})$ are $2K_2$-split minimal $(1, k)$-polar obstructions. Hence, every $(1, k)$-polar graph is a $\{K_2 \oplus 2K_2, K_1 \oplus (2K_2 + \overline{K_{k-1}})\}$-free graph.

Conversely, assume that G is a $\{K_2 \oplus 2K_2, K_1 \oplus (2K_2 + \overline{K_{k-1}})\}$-free graph. If $C = \varnothing$, then (I, S) is a $(1, 2)$-polar partition of G, so G is $(1, k)$-polar. Since G is a $K_2 \oplus 2K_2$-free graph, if $C \neq \varnothing$, then C has exactly one vertex, let say u. Let I' be the set of the neighbors of u in I. Notice that $|I'| \leq k - 2$, because G is a $K_1 \oplus (2K_2 + \overline{K_{k-1}})$-free graph. Thus, $(C \cup (I \setminus I'), S \cup I')$ is a $(1, k)$-polar partition of G. □

Now, we provide a complete characterization of $2K_2$-split graphs that admit an (s, k)-polar partition, i.e., a lemma that relates to $2K_2$-split graphs as Theorem 5 relates to pseudo-split graphs.

Lemma 18. *Let s and k be integers, $s, k \geq 2$. A strict $2K_2$-split graph $G = (C, S, I)$ is (s, k)-polar if and only if either*

1. *there is a subset C' of C such that $(S_1 \cup (C \setminus C'), S_2 \cup I \cup C')$ is an (s, k)-polar partition of G, or*
2. *there is a subset I' of I such that $(C \cup I', S \cup (I \setminus I'))$ is an (s, k)-polar partition of G,*

where S_1 and S_2 are the vertex sets of the connected components of $G[S]$.

Proof. Let (A, B) be an (s, k)-polar partition of G. If $G[S \cap B]$ is connected, then $G[S]$ inherits a T_1-partition from (A, B), so Lemma 16 implies that $I \subseteq B$. In this case, if we let $C' = C \cap B$, thus $(S_1 \cup (C \setminus C'), S_2 \cup I \cup C')$ is an (s, k)-polar partition of G. If $G[S \cap B]$ is disconnected, then Lemma 16 implies that $C \subseteq A$. In this case, if we let $I' = I \cap A$, thus $(C \cup I', S \cup (I \setminus I'))$ is an (s, k)-polar partition of G. □

The next lemma is a technical but useful consequence of Lemma 18.

Lemma 19. *Let s and k be integers, $s, k \geq 2$, and let $G = (C, S, I)$ be a strict $2K_2$-split graph. Let c and i be the cardinalities of C and I, respectively. The following statements hold true.*

1. *If both, $c \leq s-2$ and $i \leq k-1$, or both, $c \leq s$ and $i \leq k-2$, then G is an (s, k)-polar graph.*
2. *If $i \geq k$ and $c \geq s+1$, then G is not an (s, k)-polar graph.*
3. *If $c \geq s+1$ and $i \leq k-1$, then G is an (s, k)-polar graph if and only if there is a subset C' of C with at least $c-s+2$ vertices that is anticomplete to I.*
4. *If $c \leq s$ and $i \geq k$, then G is an (s, k)-polar graph if and only if there exists a subset I' of I with at least $i-k+2$ vertices that satisfies one of the following conditions:*
   (a) *I' is complete to C and $c \leq s-1$.*
   (b) *There exists a vertex $v \in C$ such that I' is complete to $C \setminus \{v\}$ and v is anticomplete to I'.*
5. *If $c \in \{s-1, s\}$ and $i = k-1$, then G is an (s, k)-polar graph if and only if at least one of the following is satisfied:*
   (a) *There exists a subset C' of C with at least $c-s+2$ vertices that is anticomplete to I.*
   (b) *There exists a nonempty subset I' of I such that satisfies one of the following conditions:*
      i. *I' is complete to C and $c = s-1$.*
      ii. *There is a vertex $v \in C$ that is completely nonadjacent to I' and such that I' is completely adjacent to $C \setminus \{v\}$.*

Proof. We only prove item 4; the other items admit similar proofs. Assume that G is an (s, k)-polar graph. Notice that, since $i \geq k$, no (s, k)-polar partition (A, B) of G is such that $I \subseteq B$. Thus, we have from Lemma 19 that there is a subset I' of I such that $(A', B') = (C \cup I', S \cup (I \setminus I'))$ is an (s, k)-polar partition of G. Since B' induces a $\overline{K_{k+1}}$-free graph, $|I \setminus I'| \leq k-2$, so $|I'| \geq i-k+2$. Additionally, since A' induces a $\overline{P_3}$-free graph, either I' is complete to C, or there is a vertex $v \in C$ such that I' is complete to $C \setminus \{v\}$ and v is anticomplete to I'. Observe that, in the case that I' is complete to C, $c \leq s-1$, because A induces a K_{s+1}-free graph. For the converse implication, we note that $(C \cup I', S \cup (I \setminus I'))$ is an (s, k)-polar partition of G. □

By using Lemma 19, it is a routine to establish the next observation. The graphs identified in it will be useful in the following section.

Remark 20. Let s and k be integers, $s \geq 2$.

1. The strict $2K_2$-split graph $G = (C, S, I)$ such that $|C| = s$, $|I| = 1$, and C is complete to I, is a minimal $(s, 2)$-polar obstruction.
2. Let $k \geq 2$. If $G = (C, S, I)$ is the strict $2K_2$-split graph such that $|C| = s+1$, $|I| = 1$ and, for two distinct vertices u and v in C, the set $C' = C \setminus \{u, v\}$ is complete to I and $\{u, v\}$ is completely nonadjacent to I, then G is a minimal (s, k)-polar obstruction.
3. Let $k \geq 3$. If $G = (C, S, I)$ is the strict $2K_2$-split graph such that $|C| = s+1$, $|I| = 1$ and, for a vertex $u \in C$, the set $C' = C \setminus \{u\}$ is complete to I and u is anticomplete to I, then G is a minimal (s, k)-polar obstruction.

Now, we present characterizations of $2K_2$-split graphs admitting (s, ∞)- and (∞, k)-polar partitions.

Lemma 21. *Let s be an integer, $s \geq 2$, and let $G = (C, S, I)$ be a strict $2K_2$-split graph. Then, G is an (s, ∞)-polar graph if and only if either $|C| \leq s$ or there is a subset C' of C with at least $|C| - s + 2$ vertices that is anticomplete to I.*

Proof. Let $c = |C|$ and $i = |I|$. Suppose that G admits an (s, ∞)-polar partition (A, B). If $c > s$, then $C \not\subseteq A$, so we have from Lemma 16 that $I \subseteq B$ and $G[S \cap A] \cong K_2$. Let $C' = C \cap B$. Since A induces a complete s-partite graph, $|C \cap A| \leq s-2$, so $|C'| \geq c-s+2$. Additionally, since $S \cap B \neq \varnothing$ and $C' \cup I \subseteq B$, we have that C' is completely nonadjacent to I, so we are done.

For the converse implication, if $s \leq c$, then $(C, S \cup I)$ is an $(s, i+2)$-polar partition of G. Otherwise, if there exists a subset C' of C having at least $c-s+2$ vertices, which is anticomplete to I, then, for any maximum clique S' of $G[S]$, $(S' \cup (C \setminus C'), (S \setminus S') \cup C' \cup I)$ is an (s, ∞)-polar partition of G. □

Unlike pseudo-split graphs, the family of $2K_2$-split graphs do not constitute a self-complementary class of graphs, so we cannot use simple arguments of complements to conclude results for (∞, k)-polarity from those of (s, ∞)-polarity on this class. In the following lemma, we characterize the $2K_2$-split graphs admitting an (∞, k)-polar partition.

Lemma 22. *Let k be an integer, $k \geq 2$, and let $G = (C, S, I)$ be a strict $2K_2$-split graph. Then, G is an (∞, k)-polar graph if and only if either $|I| \leq k-1$ or there exists a subset I' of I with at least $|I| - k + 2$ vertices such that $G[C \cup I']$ is a complete multipartite graph.*

Proof. Let $c = |C|$ and $i = |I|$. First, assume that G admits an (∞, k)-polar partition (A, B). Clearly, $S \nsubseteq A$, so $S \cap B \neq \varnothing$. From here, if $i \geq k$, then $I \nsubseteq B$, and we have from Lemma 16 that $C \subseteq A$ and $G[B \cap S]$ is disconnected. Let $I' = I \cap A$. Since B induces a K_{k+1}-free graph, $|I \cap B| \leq k-2$, so $|I'| \geq i-k+2$. In addition, $C \cup I'$ induces a complete multipartite graph because $C \cup I' \subseteq A$.

For the converse implication, let S' be a maximum clique of $G[S]$. If $i \leq k-1$, then $(C \cup S', I \cup (S \setminus S'))$ is an (∞, k)-polar partition of G. Otherwise, there is a subset I' of I with at least $i-k+2$ vertices such that $G[C \cup I']$ is a complete multipartite graph, so in this case $(C \cup I', S \cup (I \setminus I'))$ is an (∞, k)-polar partition of G. □

In the rest of the paper, we use Lemmas 19, 21 and 22 to obtain upper bounds for the order of $2K_2$-split graphs that are minimal obstructions for the properties of being (s, k)- (s, ∞)-, or (∞, k)-polar, and we develop efficient recognition algorithms for $2K_2$-split graphs with the mentioned properties.

3.1.1. Upper bounds for $2K_2$-split minimal polar obstructions

In this section, we provide upper bounds for the order of $2K_2$-split graphs that are minimal (s, k)-, (s, ∞), or (∞, k)-polar obstructions. We start with some lemmas intended to prove an upper bound for the order of $2K_2$-split minimal (s, k)-polar obstructions for integers s and k greater than or equal to 2. Observe that, for a $2K_2$-split graph $G = (C, S, I)$, if any of C, S, or I is an empty set, then G is a $(2, 2)$-polar graph. Hence, if $s, k \geq 2$, each $2K_2$-split minimal (s, k)-polar obstruction $G = (C, S, I)$ is such that C, S, and I are all of them nonempty sets. We will use this observation in what follows without any explicit mention. Next, we prove that a $2K_2$-split graph $G = (C, S, I)$ is not a minimal (s, k)-polar obstruction whenever both $|C|$ and $|I|$ are large enough.

Lemma 23. *Let s and k be integers, $s, k \geq 2$, and let $G = (C, S, I)$ be a strict $2K_2$-split graph. If $|C| \geq s+1$ and $|I| \geq k$, then G is not a minimal (s, k)-polar obstruction.*

Proof. Let $c = |C|$ and $i = |I|$. If $c \geq s+2$ or $i \geq k+1$, it follows from item 2 of Lemma 19 that there is a vertex v of G such that $G - v$ is not an (s, k)-polar graph, so G is not a minimal (s, k)-polar obstruction. The only remaining case is when $c = s+1$ and $i = k$.

By way of contradiction, assume that G is a minimal (s, k)-polar obstruction and, for an arbitrary fixed vertex $v \in C$, let $C' = C \setminus \{v\}$. Let (A, B) be an (s, k)-polar partition of $G - v$. By item 4 of Lemma 19, there exist a nonempty subset I' of I and a vertex $u \in C'$ such that I' is complete to $C' \setminus \{u\}$ and u is anticomplete to I'. But then, for any $w \in I'$, $G[C \cup S \cup \{w\}]$ is a proper induced subgraph of G that contains an induced copy of one of the minimal (s, k)-polar obstructions described in Remark 20, contradicting the minimality of G. □

Next, we prove that if $G = (C, S, I)$ is a $2K_2$-split graph such that either $|C|$ or $|I|$ is large enough, then G is not a minimal (s, k)-polar obstruction. For convenience, we separate such proof into the three following lemmas.

Lemma 24. *Let s and k be integers, $s, k \geq 2$, and let $G = (C, S, I)$ be a $2K_2$-split graph. If $|C| \geq s+2$ and $|I| \leq k-1$, then G is not a minimal (s, k)-polar obstruction.*

Proof. Let $c = |C|$ and $i = |I|$. By way of contradiction, assume that G is a minimal (s, k)-polar obstruction. Thus, for each $v \in C$, the graph $G - v$ is an (s, k)-polar graph, which implies, by item 3 of Lemma 19, that there is a subset C'_v of $C \setminus \{v\}$ with at least $c - s + 1$ vertices that is anticomplete to I. In addition, also by item 3 of Lemma 19, since G is not an (s, k)-polar graph, each vertex $v \in C$ is adjacent to at least one vertex in I.

Let H be a graph obtained from G by deleting $c - s - 1$ vertices of C. Then, H has a $2K_2$-split partition (C^*, S, I), with $|C^*| = s+1$, and such that each $v \in C^*$ has at least one neighbor in I. Therefore, the only subset C' of C^* that is anticomplete to I is the empty set, and we have from item 3 of Lemma 19 that H is not an (s, k)-polar graph, but that is absurd because H is a proper induced subgraph of G, which is a minimal (s, k)-polar obstruction. This contradiction establishes the result. □

Lemma 25. *Let s and k be integers, $s, k \geq 2$, and let $G = (C, S, I)$ be a $2K_2$-split graph. If $|C| = s$ and $|I| \geq 2k-1$, then, G is not a minimal (s, k)-polar obstruction.*

Proof. Let $c = |C|$ and $i = |I|$. If G contains a proper induced subgraph that is not (s, k)-polar, the result trivially follows, so we can assume that every vertex-deleted subgraph of G admits an (s, k)-polar partition. Thus, for any $u \in I$, $G - u$ is an (s, k)-polar graph and, by item 4 of Lemma 19, there is a subset I'_u of $I \setminus \{u\}$ with at least $i - k + 1$ vertices, and a vertex $v_u \in C$ such that, I'_u is complete to $C \setminus \{v_u\}$ and v_u is anticomplete to I'_u.

Let $x \in I'_u$. By repeating the previous argument, there is a subset I'_x of $I \setminus \{x\}$ with at least $i - k + 1$ vertices, and a vertex $v_x \in C$ such that, I'_x is complete to $C \setminus \{v_x\}$ and v_x is anticomplete to I'_x. Notice that, since $x \in I'_u \setminus I'_x$, we have that $|I'_x \cup I'_u| \geq |I'_x| + 1 \geq i - k + 2$. Additionally, $2i - 2k + 2 \geq i + 1$, because $i \geq 2k - 1$. In consequence, we have that $I'_x \cap I'_u \neq \varnothing$, otherwise

$$i = |I| \geq |I'_x| + |I'_u| \geq 2(i - k + 1) \geq i + 1,$$

which is absurd. This implies that $v_u = v_x$, because, for any $w \in I'_x \cap I'_u$, since $w \in I'_u$, the only vertex of C that is nonadjacent to w is v_u, and we know that the only vertex of C that is nonadjacent to w is v_x, because $w \in I'_x$. Therefore, $I'_x \cup I'_u$ is complete to $C \setminus \{v_u\}$, and v_u is completely nonadjacent to $I'_x \cup I'_u$. Hence, we have by item 4 of Lemma 19 that G is an (s, k)-polar graph, so it is not minimal (s, k)-polar obstruction. □

Lemma 26. *Let s and k be integers, $s, k \geq 2$, and let $G = (C, S, I)$ be a $2K_2$-split graph. If $|C| \leq s - 1$ and $|I| \geq 2k - 1$, then G is not a minimal (s, k)-polar obstruction.*

Proof. Let $c = |C|$ and $i = |I|$. If G contains a proper induced subgraph that is not (s, k)-polar, the result trivially follows. Hence, we can assume that every proper induced subgraph of G admits an (s, k)-polar partition.

Let $u \in I$. Since $G - u$ is (s, k)-polar, we have from item 4 of Lemma 19 that there is a subset I'_u of $I \setminus \{u\}$ with at least $i - k + 1$ vertices and a vertex $v_u \in C$ such that I'_u is complete to $C \setminus \{v_u\}$ and, v_u is either complete or anticomplete to I'_u. By repeating the previous argument, for any $x \in I'_u$, there exists a subset I'_x of $I \setminus \{x\}$ with at least $i - k + 1$ vertices and a vertex $v_x \in C$ such that I'_x is complete to $C \setminus \{v_x\}$ and, v_x is either complete or anticomplete to I'_x. Observe that, as it occurred in Lemma 25, since $i \geq 2k - 1$, there is a vertex $w \in I'_x \cap I'_u$.

Now, if v_u is anticomplete to I'_u, we have that v_u is not adjacent to w, so that $v_u = v_x$, and $I'_u \cup I'_x$ is a subset of I with at least $i - k + 2$ vertices that is complete to $C \setminus v_u$ and anticomplete to v_u, so we have from item 4 of Lemma 19 that G is an (s, k)-polar graph.

Since u is an arbitrary vertex in I, if the previous case does not occur, we can assume that both, v_u is complete to I'_u, and v_x is complete to I'_x. But then, $I'_x \cup I'_u$ is a subset of I with at least $i - k + 2$ vertices that is complete to C, and it follows from item 4 of Lemma 19 that G admits an (s, k)-polar partition. □

Now, we are ready to state upper bounds for the order of $2K_2$-split minimal (s, k)-, (s, ∞)-, and (∞, k)-polar obstructions.

Theorem 27. *Let s and k be integers, $s, k \geq 2$. Every $2K_2$-split minimal (s, k)-polar obstruction has order at most $s + 2k + 2$.*

Proof. Let $G = (C, S, I)$ be a $2K_2$-split minimal (s, k)-polar obstruction of order n. It follows from Lemmas 23 and 24 that, if $|C| \geq s + 1$, then $n \leq s + k + 4$. Additionally, we conclude from Lemmas 25 and 26 that $n \leq s + 2k + 2$ whenever $|C| \leq s$. Hence, we have that $n \leq \max\{s + k + 4, s + 2k + 2\}$. However, since $k \geq 2$, we have that $s + 2k + 2 \geq s + k + 4$, so the result follows. □

Theorem 28. *Let s be an integer, $s \geq 2$. Every $2K_2$-split minimal (s, ∞)-polar obstruction has order at most $2s + 4$, and this bound is tight. Consequently, there are finitely many $2K_2$-split minimal (s, ∞)-polar obstructions.*

Proof. Let $G = (C, S, I)$ be a $2K_2$-split minimal (s, ∞)-polar obstruction, and let $c = |C|$ and $i = |I|$. From Lemma 21, we have that $c > s$. In addition, since G is a minimal (s, k)-polar obstruction for some integer k, $k \geq 2$, we have from Lemmas 23 and 24, that $c \leq s + 1$, so we conclude that $c = s + 1$.

By the minimality of G, we have from Lemma 21 that, for each $u \in I$, there is a subset C'_u of C, with at least three vertices, that is anticomplete to $I \setminus \{u\}$. Additionally, since G does not admit an (s, ∞)-polar partition, Lemma 21 implies that at most two vertices of C are anticomplete to I, so each vertex $u \in I$ is adjacent to at least one vertex of C'_u. Moreover, it follows from the previous observations that, for each $u \in I$, there is at least one vertex in C'_u that is not in C'_v for any $v \in I \setminus \{u\}$. Therefore, $|\bigcup_{u \in I} C'_u| \geq i + 2$, so $c \geq i + 2$, and it follows that $|V_G| = |C| + |S| + |I| \leq 2s + 4$.

Now, for each integer $s \geq 2$, let $H_s = (C, S, I)$ be the strict $2K_2$-split graph such that $|C| = s + 1$, $|I| = s - 1$, and for an injection $f : I \to C$, the only neighbor of a vertex $v \in I$ is $f(v)$. Notice that, by Lemma 21, H_s is a minimal (s, ∞)-polar obstruction of order $2s + 4$. Hence, the bound is tight. □

Theorem 29. *Let k be an integer, $k \geq 2$. Every $2K_2$-split minimal (∞, k)-polar obstruction has order at most $2 + 2k + 2^{2k-1}$. In consequence, there is only a finite number of $2K_2$-split minimal (∞, k)-polar obstructions.*

Proof. Let $G = (C, S, I)$ be a $2K_2$-split minimal (∞, k)-polar obstruction, and let $c = |C|$ and $i = |I|$. From Lemma 22 we have that $i \geq k$. Moreover, since G is a minimal (s, k)-polar obstruction for some integer s, $s \geq 2$, we have from Lemmas 23, 25 and 26, that $i \leq 2k - 2$.

By the minimality of G, it follows from Lemma 22 that, for each vertex $x \in C$, $G - x$ is an (∞, k)-polar graph with a subset I'_x of I having at least $i - k + 2$ vertices, and such that $I'_x \cup (C \setminus \{x\})$ induces a complete multipartite graph. We claim that, for any two different vertices $u, v \in C$, if I'_v is a subset of I'_u, then the neighborhood of each vertex in I'_v is precisely $C \setminus \{u, v\}$.

To prove our claim, suppose that u and v are two distinct vertices in C such that $I'_v \subseteq I'_u$. Since $v \in C \setminus \{u\}$, v is either complete or anticomplete to I'_u. Nevertheless, G is not an (∞, k)-polar graph, so we have from Lemma 22 that $I'_v \cup C$ does not induce a complete multipartite graph, and then v is anticomplete to I'_u. Since $I'_u \cup (C \setminus \{u\})$ induces a complete multipartite graph and v is a vertex in $C \setminus \{u\}$ that is completely nonadjacent to I'_u, we have that I'_u is complete to $C \setminus \{u, v\}$.

One more time, since $u \in C \setminus \{v\}$, u is either complete or anticomplete to I'_v, but in the first case $I'_u \cup C$ would induce a complete multipartite graph, which is impossible because G is an (∞, k)-polar obstruction. Thus, u is anticomplete to I'_v, and we conclude that the neighborhood of each vertex in I'_v is precisely $C \setminus \{u, v\}$, what we previously claimed. Notice that this imply that there are not three distinct vertices $u, v, w \in C$ such that $I'_u = I'_v = I'_w$.

By our previous arguments, there are at least $\lceil c/2 \rceil$ vertices of $u \in C$ whose associated sets I'_u are pairwise different. Therefore, since $I'_u \subseteq I$, we have that $\lceil c/2 \rceil \leq |\mathcal{P}(I)| = 2^{|I|} \leq 2^{2k-2}$, from which we conclude that

$$|V_G| = |C| + |S| + |I| \leq 2 + 2k + 2^{2k-1}. \quad \square$$

3.2. Algorithms for polarity on $2K_2$-split graphs

We have observed that $2K_2$-split graphs are unipolar and co-unipolar, and hence polar graphs. In this section we also prove that the problems of deciding whether a $2K_2$-split graph is (s, ∞)-, (∞, k)- or (s, k)-polar can be efficiently solved for any choice of positive integers s and k.

We start with an observation that relates to $2K_2$-split graphs what Remark 4 relates to pseudo-split graphs. It is a direct consequence of Theorem 2, and it is key to the development of the algorithms in this section.

Remark 30. Let $G = (C, S, I)$ be a strict $2K_2$-split graph. For each $u \in V_G$, we have that $u \in C$ if and only if $\mathrm{d}(u) \geq |C| + 3$, $u \in S$ if and only if $\mathrm{d}(u) = |C| + 1$, and $u \in I$ if and only if $\mathrm{d}(u) \leq |C|$.

Moreover, $\mathrm{d}(u) = |C| + 3$ if and only if u is a vertex in C that is anticomplete to I, and $\mathrm{d}(u) = |C|$ if and only if u is a vertex in I that is complete to C.

Notice that, from the previous remark, the characterizations given in Proposition 17 imply that $(1, k)$-polarity, including $(1, \infty)$-polarity (i.e. monopolarity), can be checked in $O(|V|)$ time from the degree sequence of any $2K_2$- or C_4-split graph. Now we prove that, for any positive integer s, the $2K_2$-split graphs that admit an (s, ∞)-polar partition also can be recognized in linear time from their degree sequence.

Theorem 31. *Let s be an integer, $s \geq 2$. The problem of deciding whether a $2K_2$-split graph is (s, ∞)-polar is linear-time solvable from its degree sequence.*

Proof. Let $G = (C, S, I)$ be a $2K_2$-split graph, and let c and i be the number of vertices in C and I, respectively. If $c \leq s$, then $(C, S \cup I)$ is an (s, ∞)-polar partition of G. Otherwise, we have from Lemma 21 that G is an (s, ∞)-polar graph if and only if there exist at least $c - s + 2$ vertices of G whose degree is exactly $c + 3$. By Remark 30, these verifications can be done in linear time from the degree sequence of G. $\square$

It is not known whether (∞, k)-polarity can be decided on the class of $2K_2$-split graphs from its degree sequence. However, in the next theorem we prove that this problem can be solved in linear time.

Theorem 32. *Let k be an integer, $k \geq 2$. The problem of deciding whether a $2K_2$-split graph is (∞, k)-polar is solvable in linear time.*

Proof. Let $G = (C, S, I)$ be a $2K_2$-split graph, and let $c = |C|$ and $i = |I|$, as usual. If $i \leq k - 1$, then for any maximum clique S_1 of $G[S]$, $(C \cup S_1, I \cup (S \setminus S_1))$ is an (∞, k)-polar partition of G. Also, if the set I', consisting of all the vertices in I that are complete to C, satisfies that $|I'| \geq i - k + 2$, then $(C \cup I', S \cup (I \setminus I'))$ is an (∞, k)-polar partition of G.

Now, let us assume that $i \geq k$ and at most $i - k + 1$ vertices in I are complete to C. For each vertex $v \in C$, let I^*_v be the set of all vertices whose neighborhood is $C \setminus \{v\}$. It follows from Lemma 22 that G is an (∞, k)-polar graph if and only if I^*_v has at least $i - k + 2$ vertices for some $v \in C$.

By Remark 30, we can verify in linear time, only from the degree sequence of G, whether $i \geq k$, and whether there are $i - k + 2$ vertices of I that are complete to C. In addition, computing the cardinalities of the sets I^*_v can be done in $O(|V| + |E|)$ time in the following way. We set a counter initialized to 0 for each $v \in C$. Then, for each $w \in I$ and each $v \in N(w)$, (*i*) if $d(w) = c - 1$, we add 1 to the counter of the only non-neighbor v of w in C, and (*ii*) if $d(w) = c$, we add 1 to the counter of each $v \in C$. This way, once we finished with all vertices in I, the counter of each $v \in C$ equals $|I^*_v|$. $\square$

As a consequence of Remark 4 and Theorem 5, deciding whether a pseudo-split graph admits an (s, k)-polar partition can be done in linear time from its degree sequence. In contrast, it can be seen in Fig. 2 that, in general, it cannot be decided whether a $2K_2$-split graph is (s, k)-polar only from its degree sequence.

Despite this, in the following theorem we prove that the problem of recognizing $2K_2$-polar graphs admitting an (s, k)-polar partition is linear time solvable.

Theorem 33. *Let s and k be nonnegative integers. Deciding whether a strict $2K_2$-split graph is (s, k)-polar can be done in linear time.*

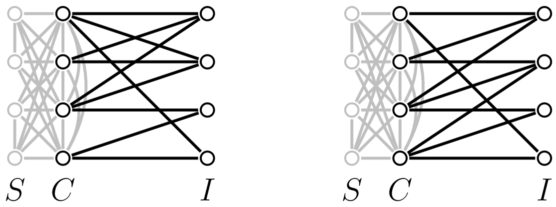

Fig. 2. Two $2K_2$-split graphs with the same degree sequence. By using Lemma 19, it can be verified that the graph on the left side is $(5, 4)$-polar but the graph on the right side is not.

Proof. Let $G = (C, S, I)$ be a strict $2K_2$-split graph, and let $c = |C|$ and $i = |I|$. Let c' be the number of vertices in G with degree exactly $c + 3$, and let i' be the number of vertices of G whose degree is c. Let I^* be the set of all vertices of G of degree $c - 1$ and, for each vertex v in C, let I^*_v be the set of all vertices $w \in I$ such that $N(w) = C \setminus \{v\}$. Observe that, proceeding similarly as we did in the proof of Theorem 32, the cardinalities of all sets I^*_v can be computed in $O(|V| + |E|)$ time.

We have the following easily-verifiable particular cases.

1. $2K_2$ is $(0, 2)$- and $(2, 1)$-polar, but it is neither $(1, 1)$- nor $(\infty, 0)$-polar.
2. $K_1 \oplus 2K_2$ is $(1, 2)$- and $(2, 1)$-polar, but it is not $(\infty, 0)$-, $(0, \infty)$-, or $(1, 1)$-polar.
3. For $c \geq 2$, $K_c \oplus 2K_2$ is $(2, 1)$-polar, but it is neither $(1, \infty)$- nor $(\infty, 0)$-polar.
4. For $c \geq 1$, $iK_1 + 2K_2$ is $(0, i + 2)$- and $(1, 2)$-polar, but it is neither $(0, i + 1)$- nor $(\infty, 1)$-polar.

These cases correspond to the conditions $C = \varnothing$ or $I = \varnothing$, and they can be checked in linear time from the degree sequence of G from Remark 30.

From the above observations, we can assume for the rest of the proof that the sets C, S and I are all nonempty. Additionally, we also consider the following special cases.

1. If $s, k \leq 1$ or $k = 0$, then G is not (s, k)-polar, because $2K_2 \leq G$.
2. If $s = 0$ and $k \geq 2$, then G is not (s, k)-polar, because $K_1 \oplus 2K_2 \leq G$.
3. If $k = 1$ and $s \geq 2$, then G is not (s, k)-polar, because $2K_2 + K_1 \leq G$.
4. If $s = 1$ and $k \geq 2$, we have from Proposition 17 that G is an (s, k)-polar graph if and only if $c = 1$ and $|\{w \in I\colon\ d(w) > 0\}| \leq k - 2$. This condition can be verified from the degree sequence of G in linear time.

If none of the cases listed before occur, then $s, k \geq 2$, so we can use the characterizations provided by Lemma 19. The following cases are based on such characterizations and Remark 30.

1. If both, $c \leq s - 2$ and $i \leq k - 1$, or both, $c \leq s$ and $i \leq k - 2$, then G is an (s, k)-polar graph.
2. If $i \geq k$ and $c \geq s + 1$, then G is not an (s, k)-polar graph.
3. If $c \geq s + 1$ and $i \leq k - 1$, then G is an (s, k)-polar graph if and only if $c' \geq c - s + 2$.
4. If $c \leq s$ and $i \geq k$, then G is an (s, k)-polar graph if and only if either
   (a) $i' \geq i - k + 2$ and $c \leq s - 1$, or
   (b) $\max_{v \in C}\{|I^*_v|\} \geq i - k + 2$.
5. If $c \in \{s - 1, s\}$ and $i = k - 1$, then G is an (s, k)-polar graph if and only if one of the following sentences is satisfied:
   (a) $c' \geq c - s + 2$.
   (b) $i' \geq 1$ and $c = s - 1$.
   (c) $\max_{v \in C}\{|I^*_v|\} \geq 1$. $\square$

4. Concluding remarks

In this work, we present a comprehensive study of polar partitions on H-split graphs for $H \in \{C_5, 2K_2, C_4\}$. We develop linear-time recognition algorithms for H-split graphs admitting monopolar, unipolar, (s, k)-, (s, ∞)-, and (∞, k)-polar partitions. This is interesting because all these properties are trivially satisfied for split graphs and, in contrast, on general graphs the recognition of monopolar and polar graphs are known to be NP-complete problems. Further, we do not know linear-time algorithms to decide (s, k)-polarity on general graphs.

We also exhibit upper bounds for the order of H-split minimal (s, k)-polar obstructions, most of them are linear in $s + k$ and tight. We use such bounds to conclude that there are finitely many H-split minimal (s, ∞)- and (∞, k)-polar obstructions, which is an unknown property on general graphs.

It is worth noting that, unlike the upper bound for the order of $2K_2$-split minimal (s, ∞)-polar obstructions provided in Theorem 28, which is linear in s and tight, the bound given in Theorem 29 for the order of $2K_2$-split minimal (∞, k)-polar obstructions is exponential in k, and we do not know whether it is tight. Regarding this observation, we pose the following question.

Problem 34. Can Lemmas 25 and 26 be improved by replacing the condition $|I| \geq 2k-1$ for a stronger one like $|I| \geq k+c_0$, for a constant c_0?

Note that, from the proof of Theorem 29, an affirmative answer to Problem 34 would imply an improvement of the bound provided in Theorem 29. Nevertheless, the next observation makes us think the answer to Problem 34 could be negative.

Remark 35. For any integers s and k, $s, k \geq 2$, the strict $2K_2$-split graph $G = (C, S, I)$ with $C = \{w\}$, $I = \{i_1, \ldots, i_{2k-2}\}$, and such that $wi_j \in E$ if and only if $1 \leq j \leq k-1$, is a minimal (s, k)-polar obstruction, and hence it is a minimal (∞, k)-polar obstruction.

We think that the next question can be answered in an affirmative way by imitating the proof of Theorem 28, which is very different from the one we used in Theorem 29.

Problem 36. Is the order of the $2K_2$-split minimal (∞, k)-polar obstructions upper bounded by a function linear in k?

Moreover, some initial explorations allow us to pose the following conjecture.

Conjecture 37. *Let k be an integer, $k \geq 3$, and let $G = (C, S, I)$ be a $2K_2$-split minimal (∞, k)-polar obstruction. Then $k \leq i \leq 2k-2$ and $c \leq 2k - i - 1$, where c and i stands for the number of vertices in C and I, respectively.*

Notice that, if the previous conjecture is true, then the order of every $2K_2$-split minimal (∞, k)-polar obstruction does not exceed $2k + 3$. In addition, such a bound would be tight since the strict $2K_2$-split graph $G = (C, S, I)$ with $C = \{c_1, \ldots, c_{k-1}\}$, $I = \{i_1, \ldots, i_k\}$ and such that, for each $j \in \{1, \ldots, k-1\}$, $N(c_j) \cap I = I \setminus \{i_j\}$, is a minimal (∞, k)-polar obstruction for every integer $k \geq 2$.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

[1] J.A. Bondy, U.S.R. Murty, Graph Theory, Springer, Berlin, 2008.
[2] Zh. A. Chernyak, A.A. Chernyak, About recognizing (α, β)-classes of polar graphs, Discrete Math. 62 (2) (1986) 133–138.
[3] R. Churchley, J. Huang, Solving partition problems with colour-bipartitions, Graphs Combin. 30 (2014) 353–364.
[4] T. Ekim, P. Hell, J. Stacho, D. de Werra, Polarity of chordal graphs, Discrete Appl. Math. 156 (13) (2008) 2469–2479.
[5] E.M. Eschen, X. Wang, Algorithms for unipolar and generalized split graphs, Discrete Appl. Math. 162 (2014) 195–201.
[6] A. Farrugia, Vertex-partitioning into fixed additive induced-hereditary properties is NP-hard, Electron. J. Comb. 11 (R46) (2004) 1.
[7] T. Feder, P. Hell, S. Klein, R. Motwani, List partitions, SIAM J. Discrete Math. 16 (3) (2003) 449–478.
[8] T. Feder, P. Hell, W. Xie, Matrix partitions with finitely many obstructions, Electron. Notes Discrete Math. 28 (2007) 371–378.
[9] S. Földes, P.L. Hammer, Split graphs, Congr. Numer. 19 (1977) 311–315.
[10] P.L. Hammer, B. Simeone, The splittance of a graph, Combinatorica 1 (1981) 275–284.
[11] V.B. Le, R. Nevries, Recognizing polar planar graphs using new results for monopolarity, in: Algorithms and Computation: 22nd International Symposium, ISAAC 2011, Yokohama, Japan, December (2011) 5-8. Proceedings 22, Springer, 2011, pp. 120–129.
[12] F. Maffray, M. Preissmann, Linear recognition of pseudo-split graphs, Discrete Appl. Math. 52 (1994) 307–312.